

PRD: Insurance Co-Pilot for Roadside Assistance

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1. Vision

A stranded driver today waits on hold, answers twenty questions, then waits again while the agent digs through their policy. The whole call takes 15-20 minutes, and the agent can help exactly one person at a time. We want that same driver talking to a voice agent within seconds, and a human agent supervising five cases at once, stepping in only where judgment is actually needed.

Success metrics. The first two are business metrics and the rest are engineering metrics. Cost per resolved case is the one we would watch most closely.

Metric	Today (assumed)	Target
Case handling time	10-12 min for this workflow (call centers average ~6 min; complex calls run past 20)	3-4 min
Concurrent cases per human agent	1 (one call at a time)	5+ at the M3 end state, when the AI runs intake, coverage and dispatch and the human approves by exception
Coverage decision accuracy	Manual, unmeasured	95%+ against an expert-labeled golden set, with near-zero false denials
Human override rate	Not tracked today	Under 5% per case type before autonomy expands
Cost per resolved case	~\$10-12 of agent time, assuming a loaded agent cost of ~\$1/min (standard call-center rule of thumb: ~\$20/hr wage x 1.5-2x for benefits, supervision and overhead)	~\$2-3 in early rollout, while a human reviews every case (2-3 min of attention each); approaching ~\$1 once routine cases auto-approve and 5x concurrency is reached. Inference adds well under \$2 per case and keeps falling

Two rules apply across these numbers. Every coverage decision must cite the policy clause it rests on; if there is no citation, the case goes to a human. And the figures are working estimates: part of M1's job is to measure them on real cases before anyone commits to them. Inference costs have been falling steadily while human costs have not, so the economics should improve over time without extra work from us. Customers also get help faster, and agents spend less time on repetitive intake and more on judgment calls.

Assumptions & constraints: browser voice only in v1 (no phone telephony); no payments/settlement, real dispatch integration, or multi-language. Call-time and cost figures above are assumptions that we validate in week one against the client's own call and cost data. Regulation is a hard constraint: every customer-facing decision needs a human approval path and a complete audit trail.

2. Key Features

There are two users here, and they want different things. The stranded driver is stressed, possibly on a dark highway shoulder, and wants to feel heard and know help is coming. The human agent needs to trust a decision in five seconds without rereading a policy. Every feature below serves one of those two.

Feature	Description
F1 Voice intake agent	Browser-based voice conversation that collects name, date of birth (to verify the policyholder and pull the right policy), car, location, damage type and situation. The bar is a natural conversation rather than a phone tree: the caller can interrupt, ramble, or answer out of order and the agent keeps up. It reads all details back for confirmation before ending. Location gets extra care (a share-GPS link via SMS rather than a spoken address), because a wrong location is the one error nothing downstream catches until a truck arrives at the wrong place.
F2 Automated coverage check	Compares the structured case against the client's policy (retrieval over policy docs). Returns covered / not covered / uncertain, with the cited clause and a confidence score.
F3 Next best action (NBA)	Recommends tow vs. repair truck from the damage type and picks the nearest garage.
F4 Agent observation dashboard	The human agent's view: live transcript, extracted fields, the coverage decision next to the exact clause it rests on, and the recommended action. Nothing reaches the customer without approval; uncertain cases are flagged for review.
F5 Client notification	SMS (simulated) with the decision and next steps, sent on human approval. SMS rather than a callback on purpose: a stressed customer wants something written down they can reread.

Useful but deferred beyond v1: photo damage assessment, phone telephony, real dispatch APIs, multi-language, and safety escalation (caller in a dangerous spot gets priority routing).

3. Prioritization Rationale

- **The dashboard (F4) carries the most weight.** Coverage denial is a regulated, high-stakes decision; we would not expect an insurer to let an AI make it unsupervised on day one, and we would not suggest it. Human approval backed by clause citations is what makes the system trustworthy enough to adopt, and we treat it as part of the design rather than a limitation.
- **F1-F3 are the thinnest end-to-end slice.** A complete flow demos better and teaches us more than depth in any single step.
- **Confidence routing is the core design decision.** The AI proceeds when it is sure and escalates when it is not, which lets speed and accuracy improve together rather than trading off against each other.

- Photo damage assessment is deliberately cut from v1: technically the riskiest piece, and the workflow proves itself without it (plan in section 6).

4. Milestones (real build, about 4 months)

Phase	Scope	Exit criteria
M1 (wk 1-4)	Co-pilot for human agents only: coverage check + NBA suggestions inside the existing workflow. No client-facing AI yet.	Agents accept 70%+ of suggestions; handling time drops
M2 (wk 5-10)	Voice intake in shadow mode first, then live for simple cases. A human approves every decision.	Intake completion and escalation rates hit agreed bars
M3 (wk 11-16)	Auto-approve for high-confidence covered cases, switched on per case type by the human agents themselves: they are accountable for the agent's decisions, so they decide when it has earned autonomy. Humans handle exceptions only. Dispatch API integration.	Override rate under 5% per case type; complete audit trail

We start with the agents rather than the customers. It is the lowest-risk place to earn trust, and every suggestion an agent accepts or overrides becomes the labeled data that justifies giving the system more autonomy later.

How the work splits: four parallel workstreams, each with a single owner: (1) voice pipeline, (2) coverage retrieval and the eval suite, (3) agent dashboard and audit trail, (4) policy ingestion and data. The eval suite is shared infrastructure from week one and everything else plugs into it.

5. Technical Risks

- **Voice reliability in the field.** Wind, traffic, stressed callers, accents. A bad transcript poisons everything downstream. Mitigation: read details back, validate fields, escalate when confidence drops.
- **Coverage correctness.** A wrong denial is a legal and reputational problem, not just a bad metric. Mitigation: clause citations, human sign-off on every denial, and a golden-case eval suite that runs on every change.
- **Latency.** Voice needs sub-second turns. The coverage check does not, so it runs after the call rather than during it.
- **Policy document mess.** Real policies are long, versioned and inconsistent. Ingestion and normalization is its own workstream, and probably the most underestimated one.
- **System failure mid-case.** If the voice agent drops mid-call or the platform is down, the case warm-transfers to the human queue with the partial transcript and extracted fields attached, so the customer never has to start over. The existing manual workflow remains the fallback path throughout rollout.
- **Compliance.** Every decision logged, explainable and reversible. EU AI Act and insurance regulators will ask.

6. AI Integration

Architecture. Production components: browser voice client → voice pipeline (STT/TTS) → orchestrator (session state, tool calls, confidence routing) → retrieval service over a normalized policy store → agent dashboard (approval queue, audit log) → notification service (SMS). The prototype collapses this into: React web app · OpenAI Realtime API voice · LLM orchestration with tool calls (coverage check, garage lookup) · synthetic policies, clients and garages in JSON/SQLite.

Model choices. Voice: OpenAI Realtime API (speech-to-speech) for the prototype, because sub-second latency and native barge-in are what make a conversation feel human, which is the thing a demo needs to show. For production I would move to a Whisper → LLM → ElevenLabs pipeline: 2-3x slower per turn, but every utterance exists as text before it is spoken, so guardrails can block a bad answer before the customer hears it, and each component can be swapped independently as better or cheaper ones ship. Reasoning (coverage, NBA): a Claude or GPT-4-class model with tool calls and retrieval over policy documents. It runs after intake, where accuracy matters and latency does not; routine extraction goes to smaller, cheaper models.

Guardrails. The voice agent collects and confirms information. It never discusses coverage and never makes promises; decisions reach the customer only through the approved SMS. No policy citation means automatic escalation, and denials or low-confidence cases always get a human. Personal data never reaches the external model raw: speech-to-text (self-hosted Whisper), name/date-of-birth extraction (small text-tagging models) and the identity check (a database match against the policy record) all run on our own infrastructure. The external reasoning model gets a redacted transcript with names, dates of birth and plates replaced by placeholders; deciding whether a scenario is covered does not require knowing who the customer is. On top of that: prompt-injection deflection, a tone filter, and a hard cap on call length that falls back to a human callback. Everything is logged: transcripts, retrievals, decisions, human actions.

Improvement loop. Every accept, override and edit on the dashboard becomes a labeled example. Labels feed a golden-case eval suite that runs on every prompt or model change; failures become few-shot examples or prompt fixes. Override rate per case type is the gate for expanding autonomy in M3. Once enough labels accumulate, fine-tune smaller models to cut cost.

Damage assessment. v1 is text only: the LLM classifies damage type and severity from the conversation and maps it to tow vs. repair. v2 adds photos via an SMS link, with a multimodal model producing a structured severity estimate plus confidence; anything uncertain goes to a human adjuster. Off-the-shelf models with few-shot examples first; fine-tuning only if accuracy demands it.